

Ideation Phase

Define the Problem Statements

Date	30 January 2026
Team ID	LTVIP2026TMIDS81120
Project Name	Smart Sorting: Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	2 Marks

Customer Problem Statement Template:

In many markets and households, people face challenges in determining whether fruits are fresh or spoiled. From a user's perspective, it is often difficult to judge fruit quality based only on visual inspection, especially when changes in color or texture are subtle. Most individuals rely on guesswork or manual checking, which can be inconsistent and may lead to purchasing or consuming spoiled produce.

As a consumer, vendor, or quality inspector, I need a simple and reliable system that can analyze fruit images and inform me whether the fruit is fresh or rotten. Without such a system, I may waste money, compromise food safety, or contribute to unnecessary food waste due to incorrect judgment.

Therefore, I require an easy-to-use fruit freshness classification application that allows me to upload an image and instantly receive a clear prediction of the fruit's condition. This helps me make informed decisions, maintain quality standards, and reduce food waste in real-world situations.

I am

quality, or a quality inspector in a food supply chain who ensures that only fresh produce reaches customers. These stakeholders are directly affected by the ability to accurately identify fruit freshness, especially in environments where manual inspection may be inconsistent or time-sensitive

I'm trying to

predict whether a fruit is fresh or spoiled based on its visual appearance so that I can make safe purchasing and consumption decisions while reducing food waste.

But

I do not have access to a reliable and easy-to-use system that can accurately analyze fruit images and clearly determine their freshness. Manual inspection is often inconsistent, and it can be difficult to judge quality when visual differences are subtle.

Because

Which makes me feel

Concerned, uncertain, and hesitant when selecting fruits, as I may not accurately identify spoiled produce, which could lead to wasted money, compromised food safety, or unnecessary food waste.